MILITARY SPECIFICATION

TRUCK, FIRE FIGHTING, STRUCTURAL, A/S32P-12;
750 GPM, 4 BY 2 CHASSIS

1. SCOPE

1.1 This specification covers a four-wheel, two-wheel drive, 750-gpm pumping capacity, fire fighting truck designated A/S32P-12.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

L-P-383	Plastic Material, Polyester Resin, Glass Fiber Base, Low Pressure Laminated
O-E-910	Extinguisher, Fire, Carbon-Dioxide (Hand and Wheeled Types)
O-F-555	Foam Liquid, Fire Extinguishing, Mechanical
VV-F-800	Fuel Oil, Diesel
ZZ-H-451	Hose, Fire, Woven-Jacketed, Rubber- or Cambric-Lined, With Couplings
ZZ-H-561	Hose, Rubber, and Hose Assemblies, Rubber, Smooth Bore, Water
GGG-A-926	Ax, (Single Bit, Double Bit, Pick Head and Mattock Head)
GGG-B-101	Bar; Chisel, Pinch, Pry, Wrecking, Digging, Tamping, Digging and Tamping; and Crowbar
GGG-C-740	Cutter, Bolt
GGG-W-665	Wrench, Spanner

Military

MIL-J-449	Jacket, Fire Hose (2-1/2-inch)
MIL-P-514	Plates, Identification, Instruction and Marking, Blank
MIL-H-4497	Hose, Cotton, Rubber Lined, Water, Braided Construction, With Couplings
MIL-M-8090	Mobility, Towed Aerospace Ground Equipment, General Requirements for

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MIL-A-8421	Air Transportability Requirements, General Specification For
MIL-S-11086	Strap, Hose, and Ladder
MIL-B-11188	Battery, Storage, Lead-Acid
MIL-S-12165	Strainer, Suction, Fire Hose, and Strainers, Suction, Hose
MIL-L-12197	Ladders, Fire, Extension; Ladders, Fire, Single; and Ladders, Fire, A-Folding Extension
MIL-N-12314	Nozzle, Fire Hose, Water, Spray and Stream
MIL-L-26598	Lantern, Electric LEU-2/E
MIL-P-43116	Pike, Wire and Pikes, Pole
MIL-T-45340	Trucks and Truck Tractors; Commercial, Gasoline and Diesel Engine Driven, 24,000 to 32,000 Pounds GVW, 4 x 2
MIL-C-52404	Connections, Hose, Fire and Water

STANDARDS

Federal

595	Colors

Military

MIL-STD-12	Abbreviations for Use on Drawings, Specifications, Standards, and Technical Documents
MIL-STD-17	Mechanical Symbols for Aeronautical, Aerospacecraft and Spacecraft Use
MIL-STD-100	Engineering Drawing Practices
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-143	Standards and Specifications, Order of Precedence for the Selection of
MIL-STD-808	Finishes, Protective, and Codes, for Finishing Schemes for Ground and Ground Support Equipment
MIL-STD-826	Electromagnetic Interference Test Requirements and Test Methods
MIL-STD-831	Test Reports, Preparation of
MIL-STD-889	Dissimilar Metals
MIL-STD-1223	Administrative Wheeled Vehicles Treatment, Painting, Undercoating, Identification Marking, Data Plates and Warranty Notice Standards
MS51317	Light, Warning, Vehicular, Rotating, D. C.

Air Force-Navy Aeronautical

AN2552	Receptacle - External Power, 28 Volt DC

PUBLICATION

Air Force Bulletin

518	Cargo Aircraft Compartment Dimensional Data

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(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Bureau of Standards

Handbook H28 Screw-Thread Standards for Federal Services

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

Standard No. 19 Motor Fire Apparatus

(Application for copies should be addressed to the National Fire Protection Association, 60 Batterymarch Street, Boston, Massachusetts 02110.)

Society of Automotive Engineers, Incorporated

SAE Standards and Recommended Practices

J534	Standard for Lubrication Fittings
J833	Male Physical Dimensions for Construction and Industrial Equipment Design

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., Two Pennsylvania Plaza, New York, New York 10001.)

3. REQUIREMENTS

3.1 Preproduction. This specification makes provisions for preproduction testing.

3.2 Components. The truck shall consist of the following major components:

Item or System Name	See Requirement
Chassis	3.7
Fire pump	3.8
Body	3.9
Auxiliary fire fighting equipment	3.10
Stored agent system	3.11
Booster hose and reels	3.12 .

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3.3 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.4 Materials. Materials shall be as specified herein. Materials not specifically covered by this or applicable specifications shall be suitable in every respect and of good quality for this type of equipment. Except where specified herein, wood shall not be used in the vehicle construction.

3.4.1 Metals. Unless otherwise specified herein, parts and components of the truck which come in contact with fire fighting agents (water, foam, foam liquid concentrate) and which come in contact with coolant liquid, except the engine, shall be fabricated of copper, brass, bronze, or 300 series stainless steel.

3.4.2 Dissimilar metals. Unless protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MIL-STD-889.

3.4.3 Protective treatment. When materials are used that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. Protective coatings that will chip, crack, or scale with age or extremes of climatic conditions shall not be used.

3.5 Design and construction. The truck shall be so constructed that parts will not work loose in service, and shall be built to withstand the strains, shocks, vibrations, and other conditions incident to operation, shipping, and storage with minimum loss of time for maintenance, repair, and servicing. The design of the truck and fire fighting equipment shall insure safe, efficient, and economical operation. All assemblies, controls, compartments, equipments, and installed attachments shall be so located that there will be no adverse interference with each other or with operations, and shall be readily accessible for maintenance, operation, removal, and replacement.

3.5.1 Functional design. The truck shall be designed to operate efficiently at military installations world wide for the purpose of providing quick response for suppression and extinguishment of real material and property fires under extreme climatic conditions. The truck shall be capable of operating on paved surface and unimproved roads as required to accomplish the fire fighting mission at such installations.

3.5.2 Reliability. The reliability of the truck to satisfy its intended use will be demonstrated by successful completion of tests specified herein.

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3.5.3 Maintainability. The truck shall be designed and constructed to provide:

a. A minimum number of parts consistent with performances required herein

b. Parts and components that are located or positioned for ease of inspection and recognition of excessive wear or potential failure

c. Ease of adjusting, servicing, and replacing parts and components

d. Use of readily available standard tools and equipments for maintenance

e. Maintenance with a minimum number of tools.

3.5.3.1 Foolproofness. Where improper installation of an item could cause malfunction of the item or the equipment in which it is installed, an unsymmetrical mounting means shall be provided where practical. That mounting shall allow the item to be installed only in its proper operating position. If an unsymmetrical mounting means is not practical, the item shall be so mounted that its proper operating position can be readily and visibly determined by service maintenance personnel.

3.5.3.2 Major assemblies. To the maximum practicable extent, it shall be possible to remove major assemblies without the necessity for draining liquids or coolants, or dismantling other assemblies. Major assemblies shall be equipped with lifting eyes or other suitable provisions for ready attachment of lifting facilities.

3.5.3.3 Cover plates. Cover plates, which must be removed for component adjustment or for component or part removal, shall be equipped with substantial quick-disconnect fastenings.

3.5.3.4 Fastening devices. All screws, pins, bolts, and similar parts shall be installed with an adequate means for preventing loss of proper tightness and adjustment. When subject to removal, replacement, or adjustment, attaching hardware for any component shall be easily removable and shall not be swagged, peened, staked, or otherwise permanently deformed.

3.5.3.5 Servicing provisions. Drains, filler plugs, grease fittings, hydraulic lines,bleeders, and checkpoints for all components shall be so located that they are readily accessible and do not require special tools for proper servicing.

3.5.4 Safety features. All space which is occupied or in which work is performed during operation, service, and maintenance of the truck shall be free of hazardous protrusions, edges, cracks, or other features that may cause injury to personnel. Safety features such as steps with antiskid tread, handrails, or guards shall be provided at all points where the protection of personnel is in question.

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3.5.4.1 Hazard guards. All rotating and reciprocating parts and all parts subject to high operational temperatures or to being electrically energized, that are of such nature or so located as to be hazardous to operating personnel, shall be insulated or guarded to the extent necessary to eliminate the hazard.

3.5.4.2 Component protection. All oil, hydraulic, and air system tubing, and all electrical lines shall be properly clipped to the frame or body structure and, except where a through-frame connector is necessary, shall be furnished with metal protective looms or grommets at each point where they pass through panel or structural members.

3.5.5 Lubrication. All parts of the truck normally requiring lubrication shall be provided with lubrication fittings conforming to SAE Standard J534. Where the use of high lubricating pressures may damage grease seals, a pressure release device shall be provided. Extended fittings shall be provided to lubricate parts or assemblies which are not readily accessible for direct lubrication, or which are likely to be overlooked because of inaccessibility.

3.5.6 Air transportability. The truck shall be designed for air transportability in accordance with MIL-A-8421 and Bulletin 518 in a Type C-130 aircraft equipped with an A/A32H-4 dual rail cargo handling system (see 6.4).

3.5.6.1 Component removal. Disconnects shall be provided for removal and reinstallation of components, such as the warning light, that may extend above the required height (see 3.16) for air transport. Removal and reinstallation of such components shall be completed by not more than one man in an elapsed time not to exceed 1 hour.

3.5.6.2 Tiedowns. The truck shall be provided with sufficient tiedown points so located that the vehicle can be restrained in all directions during air transport. Tiedowns shall be designed for vehicle restraint against forces of 9g forward, 3g upward, and 1.5g lateral. The attachment points of the truck shall be designed for use with attachment points in the aircraft which, in the expanse of aircraft floor, are of 10,000 pounds capacity and placed in a 20-inch grid pattern with 25,000-pound attachment points on approximate 40-inch centers along both sides of the aircraft floor. The tiedown points on the truck shall be designed and located on the basis that restraining chains will be at an angle of 30° with the aircraft floor. Vehicle attachment point loading capacities shall be downgraded accordingly from the standpoint of restraining horizontal, vertical, and lateral g-loads on the vehicle.

3.5.6.3 Clearance. Sufficient clearance shall be maintained around all attachment points to allow for restraining chains without interference from any other structure on the truck. The tiedown points shall have a minimum clear opening of 4 inches to allow attachment of a chain tiedown with a grabhook.

3.5.7 Weight distribution. The weight on the front axle shall be 34 ±4 percent of the weight of the truck under all loading conditions.

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3.5.8 Waterproofing. All components shall be so designed and protected that their normal functioning will not be impaired by heavy rains, road splash, formation of condensation, spillage of extinguishing agents, and leaks in the piping system.

3.6 Performance. Except as otherwise specified herein, performance requirements shall be based on operation at sea level.

3.6.1 Environmental conditions. The truck shall be capable of satisfactory performance under the following conditions without detrimental effect to subsequent operation:

a. Starting and operating in ambient temperatures ranging from -20° to +125°F

b. Starting and operating in an ambient temperature of -40°F when winterized as specified in 3.13.2.

3.6.2 Mobility. The truck, fully loaded and equipped, unless otherwise specified herein, shall be capable of the following:

a. Negotiate backward and forward, up and down, a solid ramp having sharp defined 15° angles of approach and departure, and having an incline length of not less than 1.2 times the truck wheelbase, without interference between the ramp and vehicle components. The truck shall be fully loaded except for water and foam agents

b. Maintain a speed of not less than 50 mph on level paved roads

c. Accelerate from a standstill to a speed of 35 mph in 30 seconds on level paved roads

d. Ascend and descend 30-percent grades without adversely effecting vehicle operation or causing spillage of agents, fuels, or lubricants

e. Negotiate 20-percent side slopes, in both directions, without adversely effecting vehicle operation or causing spillage of agent, fuel, or lubricants

f. Be held and controlled by the service brakes on a 30-percent grade, and be brought to a complete stop within 30 feet from a speed of 20 mph on a dry, hard surface, approximately level road free from loose material. The hand (parking) brakes shall be capable of holding the truck motionless when headed either direction on a 30-percent grade

g. Traverse paved and unimproved roads, under any load condition, as follows without adversely effecting vehicle components or installed equipments:

(1) Paved roads at an average speed of 45 mph

(2) Graded gravel roads at an average speed of 25 mph

(3) Belgian Block roads at an average speed of 20 mph.

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3.6.3 Agent discharge. The truck shall be capable of discharging water and foam (expanded solution of water plus foam liquid conforming to O-F-555) selectively from all fire pump discharge outlets. The truck shall be capable of:

a. Discharging water at a rate of 250 gpm through any three of the four 2-1/2-inch discharge outlets while pumping water at a rate of 750 gpm from either draft or hydrant supply

b. Discharging water at a rate of 250 gpm while pumping from the installed water tank supply

c. Discharging water at a rate of 60 gpm through each booster hose separately or simultaneously while pumping from draft, hydrant, or the installed water tank supply

d. Discharging foam solutions of 3 percent (3 parts foam liquid to 97 parts water) and 6 percent (6 parts foam liquid to 94 parts water) while discharging at water rates up to 200 gpm, and while pumping from draft, hydrant, or the installed water tank supply. This includes discharging at a water rate of 60 gpm through each booster hose and at higher water rates through the 2-1/2-inch discharge outlets.

3.7 Chassis. The chassis shall be the manufacturer's standard, except as modified to comply with requirements specified herein. The chassis shall be a 4 by 2 conforming to MIL-T-45340, type I, class A, group 1 or 2 as specified by the procuring activity (see 6.2). The total gross weight of the complete truck shall not exceed the manufacturer's gvw rating when fully equipped as specified herein, including all specified fire fighting equipment, serviced with fuel, lubricants, and water; filled to capacity with water in the water tank; filled to capacity with foam concentrate in the foam tank; a 1,500-pound simulated load in the hose bed; and five men weighing not less than 175 pounds each or equivalent dead weight.

3.7.1 Engine. The truck engine shall deliver a minimum of 195 net brake horsepower at governed rpm, and shall meet performance requirements specified herein. Gasoline engines shall deliver the required performance when using regular grade gasoline. Diesel engines shall deliver the required performance when using fuel conforming to VV-F-800.

3.7.1.1 Engine hourmeter. An hourmeter having a totalizing mechanism of not less than 9,999 hours shall be provided to accurately register the number of hours of engine operating time. The meter shall be of strong, rugged construction to insure continuous trouble-free performance under the most extreme equipment operating conditions specified herein. The meter shall be so mounted on the truck that it can be conveniently and easily read by personnel servicing the truck. The meter may be combined with the engine tachometer (see 3.9.2).

3.7.1.2 Cooling system. A heavy duty cooling system shall be provided.

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3.7.1.2.1 Heat exchanger. An auxiliary heat exchanger shall be provided that will permit delivery of water from the fire pump for cooling the engine liquid coolant without admixture. The heat exchanger discharge line shall be of adequate size to eliminate backpressure. The flow of water from the fire pump to the heat exchanger shall be controlled by a valve, with the valve control located at the fire pump control panel.

3.7.1.2.2 Drain cocks. Readily accessible drain cocks shall be installed at the low point of the cooling and heat exchanger systems and at any other points necessary to completely drain the systems. Drain cocks shall be of a type that will remain closed under the most severe operating conditions to be encountered by the truck.

3.7.2 Electrical system. A 12-V electrical system shall be provided. Wiring shall be weatherproof, and color coded or numbered for easy identification. Means of wiring identification shall be consistent throughout the truck. Automatic reset circuit-breaker type overload protection shall be provided.

3.7.2.1 Ignition system. A keyless ignition switch, with a red bezel light adjacent to the switch to indicate when the ignition switch is on, shall be provided. An ignition switch with soldered key shall not be provided. The truck shall be provided with a safety device to prevent engine starting unless the transmission is in neutral.

3.7.2.1.1 Alternator. An alternator shall be provided, complete with rectifier and regulator, capable of supplying 12-V, 100-amp power.

3.7.2.2 Lighting system. Clearance and side marker lights shall be provided and shall be combined. The lighting system shall be such that the headlamp circuit will not be affected by a short circuit in any of the other circuits. All metal rims, shells, mounting arms, and bases of all exterior lights shall be chrominum plated. The stoplights andtaillights shall be flush mounted with the body. All lights shall be so mounted as not to be obscured by riding personnel or mounted equipment, and to prevent damage due to operation, removal and replacement of equipment. The following additional lighting shall be furnished:

a. Two automotive-type, sealed beam spotlights, lamp No. 4435, one on each side of the windshield or near each front corner on the cab roof. Controls shall be extended inside the cab for on-off control and movement of the lamp housing through a horizontal included angle of at least 270°, and a vertical angle of at least 90° above and 35° below horizontal. The spotlight control handle shall not obscure the driver's vision, nor interfere with normal driving functions and entering and exiting the cab

b. Two adjustable, sealed-beam hose pickup lights, lamp No. 4531, mounted on the rear handrail supports. These lights shall be provided with weatherproof switches

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c. Warning signal light conforming to MS51317-2, mounted to insure its visibility 360° around the truck. The warning light shall be controlled by a switch on the cab instrument panel

d. Pump control panel lights (see 3.9.2)

e. Equipment cabinet lights (see 3.9.8).

3.7.2.3 Siren. A 12-V electronic type siren shall be provided. It shall transmit an automatic-controlled, slow-wail sound level in the magnitude of 115 to 120 dB sound pressure level, reference pressure 0.0002 microbar at a distance of 6 feet from the speaker. The siren system shall include all necessary wiring and connections so that it is adaptable to Air Force two-way radios which utilize either carbon or magnetic-type microphones, and it can be used for a public address system and for transmitting incoming radio messages through the siren speaker. Reset-type circuit breakers shall be provided in the system. A hand-operated switch for actuating the siren shall be so located on or adjacent to the cab instrument panel that it is readily accessible from both the driver and crew chief sides of the cab.

3.7.2.4 Communication equipment. When specified (see 6.2), a two-way radio communication system shall be provided. Controls of furnished equipment shall be mounted in the cab and so located as to be readily accessible from both the crew chief and driver seats. All components of the equipment shall be readily accessible for ease of maintenance, shall be protected from exposure to the weather and mechanical damage, and shall be so located or mounted as not to interfere with maintenance and operation of the truck.

3.7.2.5 Battery. The battery shall be 12-V potential conforming to MIL-B-11188, type BB-223/U(8T) and shall be furnished charged and dry, without electrolyte. The battery shall be mounted in an acid-resistant tray in an inclosed compartment ventilated to the truck exterior, and securely held in place by means of easily removable locking devices. The battery shall be readily accessible for daily inspection and service without requiring removal from the tray.

3.7.2.5.1 Receptacle. An external power receptacle in accordance with AN2552, equipped with a spring-loaded cover, shall be installed in a relatively protected position and easily accessible for connecting heavy power cables without tilting the cab. The receptacle shall be wired to provide battery charging and starting of the truck engine by an external power source. Conductors shall be sized to allow starting of the engine.

3.7.3 Fuel system. The fuel system shall be so installed to preclude the possibility of vapor lock. The system shall include an auxiliary electric fuel pump with manual control switch, and shall be interconnected in the system with separate fuel lines.

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3.7.4 Transmission. The truck shall be equipped with an automatic or semiautomatic transmission (see 6.3.1 and 6.3.2). The transmission shall be rated for heavy-duty service and shall be designed to match engine torque and speed to the load demand. Semiautomatic transmission shall be of the continuous drive, automatically stepped type. The transmission shall provide for driver selection of individual gears or ranges of gears required to meet performance requirements specified herein. The shift control shall incorporate a positive indicator for each gear or range selected. The transmission shall be so protected that the operator cannot inadvertently shift into reverse. Positive means shall be provided to prevent power transmission to vehicle drive during fire pump operations.

3.7.5 Pintle assembly. A rear pintle assembly shall be provided and so located as not to interfere with operations or with personnel standing on the rear platform.

3.7.6 Wheels, rims, tires, and tubes. Tires and tubes shall be in accordance with MIL-M-8090. Tires shall be 11.00-20, 12 ply, with highway tread. When specified (see 6.2), the tires shall be provided with nondirectional cross-country (NDCC) tread. A spare wheel and tire assembly is not required.

3.7.7 Brakes. Service brakes shall be of either the full-air or air-hydraulic type.

3.7.8 Cab. A tilt type cab shall be provided. The cab shall be of closed construction and shall provide unobstructed seating for at least three men. A rear window shall be provided, and safety glass shall be installed throughout the cab. A floormat shall be provided, and all levers and controls extending to the exterior of the cab shall be equipped with boots.

3.7.8.1 Seat belts. Seat belts shall be provided and installed for all personnel.

3.7.8.2 Cab doors. The doors shall provide easy entrance and exit of personnel with physical dimensions in the range of small to large as defined in SAE Standard J833 and wearing heavy fire fighters clothing. The outside door handles shall be readily accessible to personnel standing on the ground. Handhold bars shall be provided at each door entrance to assist personnel entering and leaving the cab, and shall be made of the same material as the body handrails (see 3.9.7).

3.7.8.3 Cab steps. The cab steps shall have nonslip steel plate tread surface and shall be within easy view and access to personnel entering and leaving the cab. The first step, upon entering the cab, shall be not more than 24 inches above ground level. The steps shall be of adequate width and depth to safely accommodate personnel wearing heavy boots.

3.7.8.4 Windshield wipers. The truck shall be equipped with electric windshield wipers.

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3.7.8.5 Buzzer. A buzzer shall be installed in the cab, and a weatherproof push-button shall be provided on one of the rear handrail supports (see 3.9.7) for signaling from the rear platform to the cab.

3.8 Fire pump. The fire pump shall be midship mounted and shall be of the centrifugal, two-stage, series-parallel type conforming to NFPA Standard No. 19. Power for driving the pump shall be transmitted from the truck engine, and controls for changing from vehicle drive to pump drive shall be located in the cab. Switching from parallel to series shall be accomplished without shutting down the pump or causing pressure surges in the hose lines. All threaded connections of the pump and pump accessories shall conform to Handbook H28. The pump shall be readily accessible for service and removal without major dismantling of other components of the truck. When drafting at a minimum lift of 10 feet through 20 feet of 4-1/2-inch hard suction hose, the pump shall deliver water capacities and corresponding discharge pressures specified below at less than the manufacturer's recommended governed speed of the truck engine:

Capacity (GPM)	Net Pump Pressure (PSI)
750	150
525	200
375	250
50	400.

3.8.1 High temperature pumping. The pump drive, gearing, and housing shall be such that, at rated capacity and pressure, the temperature rise in the gear lubricant will not exceed the maximum temperature recommended by the pump manufacturer for continuous operation in an ambient temperature of +125°F. The ambient temperature shall be measured at the intake side of the engine cooling radiator.

3.8.2 Pressure rating. The pump and all parts subject to discharge pressure, including the booster reels and hose, shall withstand a hydrostatic pressure of 500 psi for a period of 10 minutes. The suction side of the pump shall withstand a hydrostatic pressure of not less than 350 psi for a period of 10 minutes.

3.8.3 Pump drain system. A pump drain system shall be provided to drain the pump and all water-carrying lines. Drain control shall be located on the pump control panel. Drains shall discharge inside the pump compartment at a point below the bottom of the truck frame.

3.8.4 Primer pump. A 12-V electric-motor-driven, or a mechanically driven rotary primer pump shall be provided. The priming pump shall be operable with a control on the pump control panel and shall have the capacity to prime the dry fire pump through 4-1/2-inch hard suction hose at a 10-foot lift to enable the fire pump to discharge water in not more than 30 seconds under standard conditions.

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With the 4-1/2-inch hard suction hose capped, the primer pump shall be capable of developing a vacuum of 22 inches Hg at altitudes up to 1,000 feet, and shall hold the vacuum with a drop not exceeding 10 inches Hg in 10 minutes with the pump shut off. The primer pump shall be readily accessible for service and removal without major dismantling of other components of the truck.

3.8.5 Suction and discharge connections. The pump shall be provided with the following suction and discharge connections. Connections at the pump control panel shall be mounted with sufficient clearance to allow the use of spanner-type wrenches by personnel wearing arctic-type mittens. All suction and discharge connection protective caps shall be chromium-plated brass.

a. Two 4-1/2-inch NH suction connections with long handle caps. One 2-1/2-inch NH gated, gasketed female swivel suction connection with a pin or rocker lug cap. The 2-1/2-inch and one of the 4-1/2-inch connections shall be located on the left side of the truck, and the other 4-1/2-inch connection shall be located on the opposite side. Suction inlet centers shall be not more than 4 feet above ground level, and shall be provided with a readily accessible and removable brass strainer inside each inlet

b. Connection in the suction manifold for pump intake from the water tank

c. Four 2-1/2-inch NH discharge connections, two on each side of the truck with quick-closing, ball-type discharge valves and pin or rocker lug caps chained to the valve body. Discharge connections shall be provided with adapters to bring connections off at a 45° angle downward. Control for valves at the right side shall be operable from both the right and left sides of the truck. Discharge valves shall be capable of being locked in any position. Valve operation and locking and unlocking shall be a one-hand operation. Discharge outlet centers shall be as low as practicable but not more than 5 feet above ground level

d. Outlets of not less than 1-1/2-inch NPT in the discharge manifold for booster hose reels. Valve controls shall be at the pump control panel

e. Outlet of not less than 1-1/2-inch NPT in the discharge manifold for water tank refill. The valve control shall be at the pump control panel.

3.8.6 Relief valve. An automatic relief valve or pressure regulator shall be provided for controlling the pressure from all outlets. The relief valve or pressure regulator shall be so located that it is accessible for maintenance when the pump control panel or top surface of the pump housing is removed. The pressure regulator control shall be located at the pump control panel so that adjustments can be made by the operator from his normal position without the use of tools and in full view of the discharge pressure gage. The relief valve or pressure regulator shall be of such design that, when set 10 psi higher than the pump working pressure, there will not be an increase in pump pressure greater than 30 psi when discharge valves are closed slowly. The relief valve or regulator shall perform satisfactorily when pumping rated capacity at any pump working pressure over 90 psi.

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3.9 Body. The body shall be entirely separated from the cab, shall be constructed of not less than 0.1046-inch thick steel, and shall be reinforced at all points where hose and ladder clamps and hangers are attached. The body shall be free from rough and sharp edges.

3.9.1 Fire pump housing. The top surface of the pump housing shall be easily removable and shall be so constructed that, when removed, the pump can be lifted out intact without removing more than the pump suction and discharge fittings. An easily removable panel, fastened with quick-disconnects, shall be provided in the top surface to allow personnel access to the pump compartment without removing hose reels and other major components. The top surface shall be so constructed that it will support 200 pounds on any square foot of area without permanent deformation.

3.9.2 Fire pump control panel. A pump control panel shall be located on the left side of the truck. Pump pressure and suction gages connected to the pump mainfolds shall be not less than 3-1/2-inch diameter. Four 2-1/2-inch diameter discharge gages shall be connected one each to the hose line side of each discharge valve. Gages shall be of the single Bourdon type, flush mounted, provided with front removable glass and means for front external calibration, provided with devices to eliminate pressure fluctuation, and shall be moisture tight to prevent the entrance of water or fogging due to temperature change. All valve operating handles and other controls shall be operable by personnel wearing arctic-type mittens and shall be noninterfering, firmly supported, and realignable. Collars or guides shall be installed at all control openings in the panel, except openings for push-pull type controls shall be provided with rubber or nylon grommets to prevent rattle. All exposed metal parts such as valve controls, gage rings, and pressure controls shall be stainless steel, polished aluminum, or chrominum-plated materials. Hand-grip knobs on control handles may be black, temperature-resisting plastic. The panel shall be illuminated for night operations and shall include the following:

a. Pump suction gage - 30 inches Hg vacuum to 400 psi

b. Pump pressure gage - 30 inches Hg vacuum to 600 psi

c. Four discharge outlet pressure gages of 2-1/2-inch diameter

d. Two plugged 1/4-inch NPT connections for pressure test gages, one each on the pressure and the suction sides of pump

e. Pump pressure relief valve control

f. Primer pump control

g. Pump control for series and parallel operation

h. Pump drain controls

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i. Heat exchanger control

j. Engine tachometer

k. Engine oil pressure gage

l. Engine coolant temperature gage

m. Engine throttle-pull type with latch and microadjustment

n. Water tank fill control

o. Water tank discharge and drain controls

p. Water tank water level gage

q. Foam liquid tank discharge and drain controls

r. Foam liquid metering control and indicator

s. Agent system flushing control

t. Booster hose reel controls.

3.9.3 Hose bed. The hose bed shall have a capacity for stowing a minimum of 1,200 feet of 2-1/2-inch, double-jacketed, cotton, rubber-lined fire hose and 400 feet of 1-1/2-inch, double-jacketed, cotton, rubber-lined fire hose in horizontal layers. The hose bed interior shall be not less than 72 inches long, and shall be free of projections and sharp or rough edges which might chafe, tear, or hinder the placement or removal of fire hose. Reinforced hose bed divider partitions shall divide the 2-1/2-inch hose storage area in half, and shall provide a 3-1/2-inch minimum width 1-1/2-inch hose storage area on each side of the truck. Divider partitions shall be of adequate depth to completely separate the hose, and shall extend to the same height as the body side panels. Hose bed flooring shall be slatted, well-seasoned, select grade oak or other hardwood having similar characteristics. Slats shall be not more than 4-1/2-inches nor less than 3-1/2-inches wide, not less than 3/4-inch thick, carried on transverse hardwood sleepers, spaced not less than 5/8-inch nor more than 3/4-inch apart, restrained to prevent warping, and shall run the entire length of the hose bed. The trailing ends of slats shall be rounded to prevent damage to hose. The slatted floor shall be removable and shall rest on a metal floor to provide protection of the hose from road dirt and tire splash. The metal floor shall be provided with drain holes to preclude water retention, and shall support 200 pounds on any square foot of area without permanent deformation.

3.9.4 Side and rear platforms. The truck shall have side platforms extending from the rear of the cab to the front bulkhead of the forward equipment cabinets, and a rear platform extending across the full width of the truck body. The platforms shall

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be not more than 24 inches above ground level and shall have nonslip tread surfaces. The platforms shall be of sufficient size and strength to accommodate four men, weighing 175 pounds each, without noticeable deflections. All platforms shall have outer edges flanged downward, and the rear platform shall have the rear most corners radiused.

3.9.5 Toe steps. Two nonslip tread surface toe steps, one at each side of the rear of the truck body, shall be provided to permit easy access to the hose bed. Toe steps shall be of adequate width and depth to safely accommodate personnel wearing heavy boots.

3.9.6 Rear wheel housing. The rear wheel housing shall be such as to permit the use of tire chains. The housing shall be splash tight to the front, rear, and behind the wheel above the truck frame, and shall have skirting to reduce splash but permit easy removal of the wheels.

3.9.7 Handrails. The truck body shall be equipped with one rear and two side horizontal handrails, and with two rear and two side vertical handrails. Handrails shall be 1-1/4-inch minimum diameter chromium-plated brass, Chromium-plated steel, or polished stainless-steel tubing. All handrails shall be of the same material and shall be readily accessible to personnel with physical dimensions in the range of small to large as defined in SAE Standard J833. Side horizontal handrails shall be located at each side of the pump compartment (right and left sides of the truck) and shall be approximately the same length as the pump compartment. Side vertical handrails shall be located on each side of the truck and forward of the pump compartment to assist personnel stepping onto the side platforms. The rear horizontal handrail shall be not more than 6 inches forward of the rear of the hose bed and shall extend approximately the full width of the hose bed. Rear vertical handrails shall be located at each side of the rear platform. Handrails shall not interfere with operations or removal of hose and equipments.

3.9.8 Equipment cabinets. Two equipment cabinets shall be provided on each side of the truck, fore and aft of the rear wheelhousing, and shall be flush with the side platforms and wheel skirting. Side cabinet tops shall be common and shall run continuously on each side of the body. The width of the side cabinets shall be a maximum practical, and their height shall be a maximum to permit convenient mounting of the suction hose troughs and ladders. Each side equipment cabinet shall be provided with single doors, hinged on the forward side, and shall be provided with an adjustable position shelf. One equipment cabinet shall be provided at the rear of the truck beneath the hose bed, and shall be of maximum height and width practical. All cabinets shall have doors provided with recessed handles and flush-bolt-type locks, and shall be properly gasketed to prevent the entry of rain. All cabinets shall be illuminated by lights that are automatically controlled by door switches to light when the cabinet doors are opened.

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3.9.9 Mounting brackets, holders, and pads. Brackets, holders, or pads as applicable shall be provided on the truck for mounting fire fighting accessories listed in 3.10. When mounted, tools and equipment shall be readily accessible and shall present no hazard to personnel when boarding or alighting from the truck, or when removing and replacing equipment. Mounting attachments shall be of rugged construction and so designed that equipments will remain securely in place under all truck operating conditions, yet permit quick release and easy removal of equipments for service. External mounting attachments shall be chromium-plated brass, chromium-plated steel, or stainless steel.

3.10 Auxiliary fire fighting equipment. The following equipment shall be furnished and mounted on the truck:

a. Two 10-foot lengths of 4-1/2-inch hard suction hose conforming to ZZ-H-561, grade B, type II, with 4-1/2-inch NH long handle couplers on female ends and 4-1/2-inch NH rocker lug couplers on male ends. Couplers shall be chromium plated. The hoses shall be mounted in nonchafing troughs, one above the other on the left side of the truck, with female ends of hose forward but not obstructing the pump control panel. Troughs shall be fabricated of the same gage steel as that used for the truck body. The cross sectional design shall permit ventilation between the hose and trough over the entire length without chafing the hose. Quick-release fastenings shall be provided adjacent to both ends of the hose and shall be of the heavy-duty, cam-locking type so designed as to permit simultaneous release of one end of all three hoses through a single operation of the locking lever. The fastener shall not permit movement of the hose during travel of the truck

b. One 16-foot length of 4-1/2-inch soft suction hose conforming to ZZ-H-451, type III, class B with 4-1/2-inch NH long handle, female, gasketed coupling on both ends. Couplings shall be chromium plated

c. Ladders shall be securely mounted on the right side of the truck body above the equipment cabinets and wheel housing, with quick-release fastenings. Ladder position shall not interfere with removal of booster hose or with operation of valves and connections. Ladders shall conform to MIL-L-12197, aluminum alloy, as follows:

(1) One 36-foot extension ladder, class 1, style B

(2) One 14-foot roof ladder, class 2, with folding hooks

(3) One 21-foot A-folding extension ladder, class 3, style B

d. Two carbon-dioxide (CO_2) extinguishers conforming to O-E-910, type I, size 15

e. One pike pole conforming to MIL-P-43116, type III, style B, size 1

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f. One crowbar conforming to GGG-B-101, type II, class 1, size 4

g. One wrecking bar conforming to GGG-B-101, type V, class 1, style B

h. One pick head fire ax conforming to GGG-A-926, type II

i. One chopping ax conforming to GGG-A-926, type I, class 1, design A, 5-pound head

j. Two universal spanner wrenches conforming to GGG-W-665, type V

k. Two adjustable combination hydrant and spanner wrenches conforming to GGG-W-665, type IV

l. Two double-eye wrenches conforming to GGG-W-665, type VI

m. One bolt cutter conforming to GGG-C-740, type II, class 3, 5/8-inch rod

n. Six hose and ladder straps conforming to MIL-S-11086

o. One lever-operated hose clamp for 2-1/2-inch single jacketed hose

p. One hose jacket conforming to MIL-J-449

q. Two lanterns, less dry cells and with mounting bracket, conforming to MIL-L-26598, class 2

r. One strainer conforming to MIL-S-12165, type I, with 4-1/2-inch NH internal coupling, brass or chromium-plated brass

s. Two nozzles, 1-1/2-inch NH combination spray and stream conforming to MIL-N-12314, type II, class A, chromium plated

t. Two nozzles, 2-1/2-inch combination spray and stream conforming to MIL-N-12314, type II, class A, chromium plated

u. Hose connections conforming to MIL-C-52404, brass or chromium-plated brass, as follows:

(1) Two nipples, 1-1/2-inch NH external, type XIX

(2) Two nipples, 2-1/2-inch NH external, type XIX

(3) One siamese connection, 2-1/2-inch NH internal swivel by two gated 1-1/2-inch NH externals, type XIV, class B

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(4) One siamese connection, 2-1/2-inch NH external by two gated 2-1/2-inch NH internal swivels, type XIV, class E

(5) Two reducers, 2-1/2-inch NH internal by 1-1/2-inch NH external, type XV, class C

(6) Two couplings, 1-1/2-inch NH internal, type XVII, class A

(7) Two couplings, 2-1/2-inch NH internal, type XVII, class A

(8) One coupling, gated, 2-1/2-inch NH internal by 2-1/2-inch NH external, type XVII, class C.

3.11 Stored agent system. The stored agent system shall include a foam proportioner, agent (water and foam liquid) tanks, piping, and all agent fill, drain, and flushing provisions associated therewith.

3.11.1 Agent tanks. The agent tanks shall be so installed that they can be readily removed from the truck by no more than disconnecting piping to the tanks, removal of bolt-on outer truck sheet metal, and unbolting the tank mountings. The tanks shall be provided with removable top covers to allow access for cleaning and repair of all internal portions of the tanks. Top covers shall be sealed to prevent agent leakage under all operating conditions, and shall support not less than 200 pounds on any square foot of surface area. Top retaining fasteners shall not come in contact with agents and shall be replaceable. Baffle partitions shall be installed in the tanks to prevent longitudinal surges or side-sway of the truck while the truck is maneuvering with the tanks filled or partially filled to any level. Baffles in the foam liquid tank shall also prevent excessive foaming of the liquid. Baffles and other tank interior structure shall prevent vortices during pumping. Connections to the tanks shall be reinforced and provided with flexible joints or hose to allow for motion of the tanks and piping. The tanks shall be fabricated of 300 series stainless steel. High quality fiberglass reinforced plastic materials may be used in fabricating the tanks when approved by the procuring activity. Plastic materials shall conform to L-P-383, and the process specification shall be submitted for approval. The tanks shall withstand an internal static pressure of 5 psi without leaking, and shall be so designed that at least 99 percent of their agents will be discharged during pumping operations with the truck positioned on level ground.

3.11.1.1 Water tank. The water tank capacity shall be not less than 300 gallons. The tank shall be provided with a hinged or easily removable, gasketed, inspection cover of at least 15 inches square or diameter.

3.11.1.1.1 Water tank vent. The tank shall be provided with nonsplash combination overflow and vent of not less than 3 inches nominal pipe size extending below the truck frame and directed toward the ground.

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3.11.1.1.2 Water tank top fill. The tank shall be provided with a top fill with a gasketed, hinged cover or latch. The fill shall be so designed that a 2-1/2-inch hose with coupling can be placed in the opening and the tank filled without the hose being held by the operator. The fill opening shall include a rigidly constructed removable strainer, of at least 10 mesh, to prevent entry of foreign matter into the tank during fill operations.

3.11.1.1.3 Water tank drain and discharge. The tank discharge shall be from the lowest point in the tank to provide complete drainage. The discharge (pump suction) line shall be provided with a quick-opening shutoff valve, and a drain valve at the lowest point in the line to provide complete drainage. Valve controls shall be operable from the pump control panel. A check valve shall be provided to prevent entry of foam solution into the tank.

3.11.1.2 Foam liquid tank. The foam liquid tank capacity shall be not less than 55 gallons.

3.11.1.2.1 Foam liquid tank vent. The tank shall be provided with a splashproof vent directed toward the ground.

3.11.1.2.2 Foam liquid tank top fill. The tank shall be provided with a filler neck, incorporating a removable strainer and a hardened steel cutting edge can puncturing device, so that the tank can be easily filled from standard 5-gallon foam liquid containers. The filler neck shall be located in the left forward corner of the tank, extended far enough into the tank to minimize foaming while filling, and equipped with a gasketed-type cover.

3.11.1.2.3 Foam liquid tank drain and discharge. The tank drain and discharge shall be from the lowest point in the tank to provide for complete drainage. The tank shall be provided with a drain with discharge at a point below the bottom of the truck frame, and shall be readily accessible from one side of the truck. The drain valve shall be located immediately adjacent to the tank to minimize the length of piping between the tank and valve. Valve controls shall be operable from the pump control panel.

3.11.2 Foam proportioner. The foam proportioner shall be of the around-the-pump type. Mechanical foam liquids conforming to O-F-555 shall be inducted by venturi action into a line carrying a portion of the solution from the fire pump discharge. The foam liquid-water mixture shall be drawn into the pump intake and shall mix with the main water flow. A metering valve shall be provided in the foam liquid line leading into the venturi for controlling the amount of foam liquid entering the venturi. A quarter-turn ball valve shall be installed between the foam liquid tank and the metering valve, and between the pump discharge and venturi. A check valve shall be installed between the metering valve and venturi to prevent flow of water into the foam liquid tank. The foam liquid metering shall be calibrated to regulate the proportioning of foam liquid to total solution flow. The regulator shall be provided with a

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graduated indicator to show the discharge percentage of foam. The indicator shall not be on the bottom of the foam liquid metering valve. The system shall deliver 3 to 8 percent foam solution through the normal operating pressure range of the fire pump and at discharge capacities up to 200 gpm. Means shall be provided for complete flushing of foam residue from the system. The indicator, metering, shutoff, and flushing controls shall be operable from the pump control panel.

3.12 Booster hose and reels. The truck shall be equipped with two booster hose reels, one on each side of the truck mounted above the pump compartment. Four 50-foot lengths of 1-inch inside diameter, smooth bore, rubber-lined hose conforming to MIL-H-4497 shall be installed on each reel. A 1-inch combination spray and stream handline nozzle conforming to MIL-N-12314, type II, class A, chromium plated shall be connected to the ends of the booster hoses. Nozzles shall be securely mounted in quick-release devices. Hose guides or rollers shall be provided at any location where the truck body or mounted equipment may offer interference with operation of the hose. Hose shall be connected to the pump discharge manifold with a 1-1/2-inch minimum diameter pipeline having a flexible section to take up vibration and shall incorporate shutoff valves and drain valves, operable from the pump control panel, to effect complete draining of the system. Hose reels shall have a handcrank manual rewind and shall be equipped with a friction brake payout to prevent the reels from turning due to backlash and pressure surges, and to prevent overtravel. Hose reels shall be readily accessible to personnel, with physical dimensions in the range of small to large as defined in SAE Standard J833, while standing on the side platforms.

3.13 Low temperature provisions

3.13.1 Winterization for -20°F operation. The truck shall be capable of operating in ambient temperatures as low as -20°F. Group 1 (gasoline engine) trucks shall be capable of engine starts in ambient temperatures as low as -20°F without benefit of other than standard engine accessories. Group 2 (diesel engine) trucks shall be capable of engine starts in ambient temperatures as low as 0°F, and shall be equipped with thermostatically controlled radiator shutters, an ether injection system, and other provisions required to meet this performance. All trucks shall be so designed that additional features required for winterization to -40°F, as described in 3.13.2, may be readily installed. To meet -20°F operational requirements, at least the following minimum provisions shall be incorporated in the truck:

a. A fully inclosed fire pump compartment

b. A minimum 1-inch thickness of expanded polyethylenechloride or polyvinylchloride insulation of the water tank, foam liquid tank, and fire pump compartment.

3.13.2 Winterization for -40°F operation. When specified (see 6.2), a winterization system shall be provided, in addition to that described in 3.13.1, that will permit satisfactory operation of the truck in ambient temperatures as low as -40°F. The

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winterization system shall be powered by the truck electrical system. Automatic switches shall be provided to shutoff the system in the event of ignition or overheating failures. The on-off control switch, and a red bezel light to indicate when the system is operating, shall be installed on the cab instrument panel. The system shall be so designed and installed that major components of the system can be removed and reinstalled without draining coolant from the system and without loss of coolant in excess of the capacity of the component being removed. To meet the -40°F operational requirements, at least the following minimum provisions shall be incorporated in the truck:

a. A combustion, liquid coolant circulating, automatic cycling type heater for heating the water and foam liquid agents, the fire pump compartment, and hose reels

b. An insulated inclosure around each hose reel. The inclosures shall be such that they can be opened on the outboard sides of the truck, allow for unobstructed withdrawal of booster hose from the reels, and not interfere with visibility of instruments and operation of controls

c. Thermostatically controlled radiator shutters on both group 1 and group 2 trucks.

3.14 Part numbering and interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of MIL-STD-100 shall govern the manufacturer's part numbers and changes thereto.

3.15 Electromagnetic interference. The truck shall comply with the requirements of MIL-STD-826, method 7001. This requirement shall apply to the truck, including any and all equipment used in or on the truck, during operation under all conditions specified herein.

3.16 Dimensions. The truck shall conform to the following dimensions:

a. Width - 96 inches maximum

b. Height - The overall height of the truck in a fully operational configuration, except for water and foam liquid agents, shall not exceed 104 inches. The overall height may be reducible as required for air transport (see 3.5.6.1)

c. Length - Minimum practicable

d. Cab dimension - The distance from the centerline of the front axle to the rear of the cab shall not exceed 30 inches.

3.17 Finishes and protective coatings. The truck chassis shall be cleaned, treated, and painted in accordance with best commercial practice. The truck body and all equipment not covered above shall be cleaned and treated in accordance with

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MIL-STD-808. Paint colors shall conform to Standard 595. Cab and body exterior surfaces shall be red, color 11136. Cab and equipment compartment interior surfaces shall be green, color 24300. Side and rear platform, and step surfaces shall be color 17178.

3.17.1 Undercoating. When specified (see 6.2), the truck shall be undercoated in accordance with MIL-STD-1223.

3.18 Marking

3.18.1 Lettering and numbering. All lettering and numbering shall be reflectorized tape as specified in MIL-STD-808 and by the procuring activity (see 6.2).

3.18.1.1 Tire pressure. The recommended tire inflation pressure shall be stenciled on the truck as near to each wheel as practicable.

3.18.2 Data plates. Instruction and identification data shall be permanently and legibly etched upon metal plates in accordance with MIL-P-514. All plates shall be of a size and shape consistent with information required thereon. Abbreviations shall be in accordance with MIL-STD-12. Symbols shall be in accordance with MIL-STD-17.

3.18.2.1 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130. A nameplate shall be mounted in the cab, and except for serial number and date of delivery which may be stamped, all data shall be inscribed. The following nomenclature shall be added to the nameplate:

Truck, Fire Fighting A/S32P-12, 750 GPM, 4 By 2
Water Capacity_____ Gallons
Foam Concentrate Capacity _____Gallons
Vehicle Weight Unloaded _____ Pounds
Gross Vehicle Weight _____ Pounds
Fuel Grade _____
Oil, Engine, Above 32°F, SAE Grade _____
Oil, Engine, Below 32°F, SAE Grade _____
Date of Delivery _____.

3.18.2.2 Identification of controls. All valves, gages, controls, and indicators, other than standard vehicular instrumentation, used in operation of the truck shall be identified by securely attached nameplates of such composition that exposure to oil, dirt, light, et cetera, will not cause them to fade or become eradicated. Tags and decals shall not be used.

3.18.2.3 Operating instructions. Brief operating and precautionary instructions shall be permanently affixed near the controls and switches. Instructions shall be clear, concise, and adequate to enable operation without damage to the equipment or injury to personnel.

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3.18.2.3.1 Caution plate. The following caution plate shall be installed at a conspicuous location on the cab dash panel.

CAUTION

THIS TRUCK IS EQUIPPED WITH RADIO INTERFERENCE SUPPRESSION DEVICES. CARE SHOULD BE EXERCISED DURING TRUCK MAINTENANCE TO RETAIN AND REINSTALL THESE DEVICES PRIOR TO REUSE OF THE TRUCK.

3.18.2.3.2 Flushing instruction plate. An instruction plate shall be installed on the pump control panel and shall contain specific data concerning flushing of the foam/water system piping.

3.18.2.3.3 Lubrication instruction plate. A lubrication plate shall be provided directing attention to all lubrication fittings and shall specify the range and grade of lubricant required for operational temperatures. The plate shall be permanently attached to the truck in an accessible and convenient location.

3.18.2.4 Pump delivery plate. A plate shall be installed at the pump control panel indicating the fire pump water delivery rate in gpm at 150, 200, 250, and 400 psi pressures with the truck manufacturer's specified engine speeds for these delivery rates.

3.18.2.5 Transportation data plate. A transportation data plate shall contain at least the following:

a. Side and rear silhouette views of the truck

b. Horizontal and vertical location of the truck center-of-gravity marked on the silhouette views

c. Shipping weight

d. Loading cubage

e. Overall length, width, and height.

3.19 Workmanship. The truck, including all parts and accessories, shall be fabricated and finished in a thoroughly workmanlike manner. Particular attention shall be given to freedom from blemishes, defects, burrs and sharp edges; marking of parts and assemblies; thoroughness of soldering, welding, brazing, riveting, painting, and wiring, alignment of parts and tightness of assembly screws, bolts, et cetera.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection

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requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing of the truck shall be classified as follows:

a. Preproduction testing . . . . . See 4.4

b. Acceptance tests . . . . . See 4.5.

4.3 Test conditions

4.3.1 Preparation for test. All trucks submitted for test shall contain all specified components and shall be completely assembled and properly prepared for operation. The truck, including all components, shall be correctly adjusted, lubricated, and serviced; and all equipment which has been ordered shall be mounted for immediate operation. Fuel and agent tanks shall be full unless otherwise specified for a particular test.

4.3.2 Data. During the tests specified herein, at least the following data, as applicable, shall be recorded at intervals not greater than 15 minutes or as otherwise indicated. Additional data may be required and recording intervals decreased for tests of short duration or as otherwise dictated by test objectives.

a. Time started and completed

b. Ambient temperature and atmospheric pressure

c. Agent level in water and foam liquid tanks

d. Fire pump inlet pressure

e. Fire pump discharge pressure

f. Agent flow rate through each hose line

g. Engine speed (rpm)

h. Engine oil temperature

i. Engine coolant temperature.

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4.3.3 Test observations. Throughout all tests specified herein, the truck shall be closely observed for the following conditions which shall be considered cause for rejection.

a. Failure to conform to design and performance requirements specified herein

b. Spillage or leakage of any liquids including fire fighting agents, coolants, fuels, and lubricants under all operating conditions

c. Structural failure of any part including permanent distortion, or evidence of impending failure

d. Evidence of excessive wear

e. Interference between truck components or between the truck and the ground and all required obstacles, with the exception of normal contact by the tires

f. Misalignment of parts and components

g. Instability in agent system pressure and flow

h. Evidence of undesirable roadability features including instability in handling during braking, turning, and negotiating slopes

i. Conditions which present a safety hazard to personnel during operation, service, and maintenance

j. Overheating of the truck engine or any component of the truck.

4.4 Preproduction testing (see 6.2)

4.4.1 Test sample. One truck shall be subjected to the tests specified in 4.4.3.

4.4.2 Test report. After completion of the preproduction tests, a test report shall be prepared in accordance with MIL-STD-831, and three complete copies of the report furnished to the procuring activity.

4.4.3 Preproduction tests. The preproduction tests shall consist of all tests specified under 4.6.

4.5 Acceptance tests. The acceptance tests shall consist of the following:

a. Individual tests . . . . . See 4.5.1

b. Sampling plan and tests . . . . . See 4.5.2.

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4.5.1 Individual tests. Each truck shall be subjected to the following tests as described under 4.6:

a. Examination of product . . . . . See 4.6.1

b. Mechanical operation . . . . . See 4.6.2

c. Agent tanks, static pressure test . . . . . See 4.6.3

d. Agent system, hydrostatic pressure test . . . . . See 4.6.4.

4.5.2 Sampling plan and tests. One truck shall be selected at random from each fifty, or fraction thereof produced, and subjected to the one-hundred-mile road test as described in 4.6.5.

4.5.2.1 Rejection and retest. When an item selected from a production run fails to meet the specification, no items still on hand or later produced shall be accepted until the extent and cause of failure have been determined and appropriately corrected. The contractor shall explain to the Government representative the cause of failure and the action taken to preclude recurrence. After correction, all tests shall be repeated.

4.5.2.2 Individual tests may continue. For production reasons, individual tests may be continued pending the investigation of a test failure. Final acceptance of the items on hand or produced later shall not be made until it is determined that all items meet all the requirements of the specification.

4.5.3 Defects in items already accepted. The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and methods of correcting them.

4.6 Test methods

4.6.1 Examination of product. The truck shall be inspected to determine compliance with requirements of this specification with respect to materials, dimensions, workmanship, and markings.

4.6.2 Mechanical operation. A thorough inspection shall be made of the truck to insure proper functioning of all mechanical and electrical components, accessibility of all controls and levers, and safe operation. All components shall be moved through their maximum normal displacement applicable to operations of driving and agent discharge, and shall exhibit no malfunctioning, leaks, objectional rough operation, or irregular movements.

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4.6.3 Agent tanks, static pressure test. The water and foam liquid tanks shall be subjected to an internal static air pressure of 5 psi for not less than 30 minutes. A soap solution shall be sprayed on the exterior of the tanks to facilitate detection of leaks that may exist.

4.6.4 Agent system, hydrostatic pressure test. The agent system piping, hose reels and hose, and all components on the discharge side of the pump shall be subjected to a hydrostatic pressure of 500 psi for a period of 10 minutes. The agent system piping and all components on the suction side of the pump, from the tank shutoff valves, shall be subjected to a hydrostatic pressure of 350 psi for a period of 10 minutes.

4.6.5 One-hundred-mile road test. The fully loaded truck shall be operated over hard surface roads for a distance of 100 miles at an average speed of 35 mph. During the test, the following functions shall be performed:

a. Normal or average stops, 10 times

b. Quick stops from a speed of at least 20 mph by instantaneous application of full service braking power, 5 times

c. Sharp right and sharp left turns, 5 times each

d. Operation through all forward and reverse gears.

4.6.6 Ramp test. The truck, fully loaded except for water and foam liquid agents, shall be driven forward from a solid horizontal surface and ascend a solid ramp equal in length to at least 1.2 times the truck wheelbase and making an angle of 15° with the horizontal. It shall proceed across the ramp crest onto an upper connecting horizontal surface until all wheels are resting on the upper surface. The truck shall then be backed from the upper horizontal surface, down the ramp and onto the lower horizontal surface. The truck shall then be turned around and backed up the ramp onto the upper horizontal surface and then driven forward down the ramp and onto the lower horizontal surface.

4.6.7 Pump housing top surface removal. It shall be demonstrated that the top surface of the fire pump housing can be easily removed and provide adequate access for removal of the fire pump. It shall also be demonstrated that there is provision in the top surface for personnel access to the fire pump without requiring removal of hose reels and other major components.

4.6.8 Top surface loading. It shall be demonstrated that the top surface of the truck body, extending from the forward bulkhead of the fire pump housing to the rear of the hose bed, will withstand 200 pounds on any square foot of area without permanent distortion, damage, or evidence of lack of rigidity.

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4.6.9 Platform loading. It shall be demonstrated that the rear and side platforms will each accommodate and safely support four men, weighing a minimum of 175 pounds each, without damage or noticeable deflection.

4.6.10 Agent tank capacities. The water tank and the foam liquid tank shall be filled to capacity, and the quantity of agent required to fill each tank shall be accurately measured. It shall be demonstrated that the water tank can be filled from the connection on the left side of the truck and from the top fill using 2-1/2-inch hose, and that the foam liquid tank can be easily filled from the top fill using 5-gallon foam liquid containers and the installed puncturing device.

4.6.11 Hose reel operation. It shall be demonstrated that the booster hose can be easily withdrawn from each reel at various angles away from the truck without undue operational interference, and that positive means are provided to prevent hose overtravel.

4.6.12 Pumping tests. The following tests of the completely assembled truck shall be conducted and data recorded in accordance with NFPA Standard No. 19 and as specified herein.

4.6.12.1 Run-in test. The truck shall be subjected to a run-in test of not less than 4 continuous hours while drafting water at a 10-foot lift through 20 feet of 4-1/2-inch hard suction hose. The fire pump shall be completely drained at the beginning of this test, and the time for the primer pump to initiate flow shall be recorded. During this test, the fire pump shall discharge water as follows:

a. 750 gpm at 150 psi for 1 hour continuous pumping

b. 525 gpm at 200 psi for 1/2 hour continuous pumping

c. 375 gpm at 250 psi for 1/2 hour continuous pumping

d. 50 gpm at 400 psi for 15 minutes continuous pumping.

4.6.12.2 Primer pump vacuum test. The primer pump, using a capped 20-foot section of 4-1/2-inch hard suction hose, shall be tested for vacuum developed at altitudes up to 1,000 feet and vacuum drop over a period of 10 minutes.

4.6.12.3 Pressure control test. With the pressure relief valve or regulator set at 10 psi higher than the fire pump working pressure, all discharge outlets shall be slowly closed while discharging rated capacities at 90, 150, 200, 250, and 400 psi, and the discharge pressure increase recorded under each condition.

4.6.12.4 High temperature pumping test. The properly instrumented truck shall be placed in a hot chamber and subjected to an ambient temperature of 125° ±2°F for a period of at least 24 hours. Upon completion of the 24-hour period and while

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maintaining the 125°F temperature, the truck engine shall be started and operated for 3 continuous hours while drafting water at a 10-foot lift through 20 feet of 4-1/2-inch hard suction hose. The fire pump shall be completely drained at the beginning of this test, and the time for the primer pump to initiate flow shall be recorded. During this test, truck engine and fire pump temperatures shall be carefully monitored, and the fire pump shall discharge water as follows:

a. 750 gpm at 150 psi for 2 hours continuous pumping

b. 525 gpm at 200 psi for 1/2 hour continuous pumping

c. 375 gpm at 250 psi for 1/2 hour continuous pumping.

4.6.13 Agent discharge test. Water and foam agents shall be discharged from the truck to determine that:

a. Water discharge rates of 250 gpm can be obtained through any three of the four 2-1/2-inch discharge outlets while pumping water at a rate of 750 gpm from draft and hydrant supply

b. A water discharge rate of 250 gpm can be obtained while pumping from the installed water tank supply

c. A water discharge rate of 60 gpm can be obtained through each booster hose separately or simultaneously while pumping from draft, hydrant, or the installed water tank supply

d. With corresponding settings of the foam liquid metering valve, and while pumping from draft, hydrant, or the installed water tank supply, 3 percent and 6 percent foam solutions can be obtained while:

(1) Discharging at a water rate of 60 gpm through each booster hose separately or simultaneously

(2) Discharging at a water rate of 200 gpm through the 2-1/2-inch discharge outlets.

4.6.14 Agent tank exhaustion. It shall be demonstrated that at least 99 percent of the water tank and foam liquid tank capacities can be discharged during pumping operations while the truck is positioned on level ground.

4.6.15 Low temperature operation. The truck shall be serviced with low temperature lubricants and coolants; instrumented for temperature readings at the truck

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engine, fire pump compartment and associated piping, water tank interior, foam liquid tank interior, and hose reels; and shall be placed in a cold room and subjected to the following conditions and tests:

a. The truck, with dry fire pump and empty agent tanks, shall be subjected to an ambient temperature of -20° ±2°F (0° ±2°F for group 2 trucks) for a period of at least 24 hours. At the end of this period, it shall be demonstrated that the truck engine can be started without assist from heating aids or external power source

b. Upon completion of the engine start test, and with the engine running, the water tank and foam liquid tank shall be filled to capacity with their respective agents. Temperatures of the agents used in filling the tanks shall not exceed 60°F. With the agent tanks filled, the truck (both groups 1 and 2) shall be subjected to an ambient temperature of -20° ±2°F for a period of at least 2 hours. At the conclusion of the 2-hour period, the fire pump shall be operated and the capability to discharge water and foam agents shall be demonstrated by operating each 2-1/2-inch discharge valve. There shall be no evidence of freezing in the agent system

c. When winterized for -40°F operation, the truck with engine running, winterization system operating, agent tanks filled, and with a wet fire pump shall be subjected to an ambient temperature of -40° ±2°F for a period of at least 4 hours. At the conclusion of the 4-hour period, the fire pump shall be operated and the capability to discharge water and foam agents shall be demonstrated by operating each 2-1/2-inch discharge valve. It shall be demonstrated that the booster hose can be withdrawn from each reel and replaced without difficulty. There shall be no evidence of freezing in the agent system.

4.6.16 Air transportability test. An engineering evaluation shall be conducted on the truck to determine compliance with MIL-A-8421 tests for air transport in a C-130 aircraft. The evaluation shall include all major components and their capability to withstand the 9g forward load as well as g-loadings specified in MIL-A-8421. The evaluation shall include tiedown fittings.

4.6.16.1 Configuration test. If it is required that certain truck components be removed for air transport, it shall be demonstrated that such components can be easily removed and the reduced height for air transport shall be measured. The time required for one man to remove and replace the component(s) shall be determined.

4.6.17 Electromagnetic interference. The truck shall be subjected to the radio interference suppression tests in accordance with MIL-STD-826, method 7001 when any and all equipment used in or on the truck is operating.

4.6.18 Mechanical checks. A critical mechanical inspection shall be made of all components to determine their operability and any damage or undue wear incurred during preproduction testing. Teardown and parts measurement shall be made in those cases where service life is in question.

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4.6.19 Truck mobility and performance tests. The following tests shall be conducted at the Aberdeen Proving Ground, Maryland.

4.6.19.1 Mechanical inspection. A critical mechanical inspection shall be made of all truck and fire fighting equipment components, and data recorded as to their condition, defects in manufacture, damage in transit, and damage through use prior to test.

4.6.19.2 Dimensions. Measurements of the truck overall length, width, height, rear of cab to centerline of front axle, wheelbase, and ground clearance shall be made.

4.6.19.3 Weight. The curb and gross weights of the truck shall be determined. Allowances for driver and crew shall be 175 pounds per man.

4.6.19.4 Load distribution. The load on each wheel and on each axle of the truck at curb and gross weights shall be determined.

4.6.19.5 Braking. The truck, with gross load, shall be brought to a stop using the service brakes when headed both up and down a dry, hard-surface, 30-percent grade free of loose material. The parking brake shall be applied, and the service brakes released. The truck shall remain stationary. The truck shall be driven at a speed of 20 mph on a level, hard-surface road free of loose material, and the distance required to bring the truck to a stop by applying the service brakes shall be determined. Appropriate methods shall be utilized to consider driver reaction time. There shall be no evidence of harsh operation during brake applications.

4.6.19.6 Maximum speed. The truck, with gross load, shall be operated on a level paved course to determine the maximum vehicle speed that can be obtained. The speedometer shall be calibrated to determine true road speed.

4.6.19.7 Acceleration. The truck, with gross load and engine idling, shall be accelerated from a standstill to 35 mph on a level paved course. The time-speed relationship shall be determined. The speedometer shall be calibrated to determine true road speed.

4.6.19.8 Gradeability. The truck, with gross load, shall be driven up and down a 30-percent grade. The ability of the truck to negotiate this grade, and the ability to start and idle the truck engine while the truck is in both ascending and descending positions shall be determined. Engine oil and fuel pressures shall be noted. There shall be no evidence of loss of fuel, lubricants, or fire fighting agents.

4.6.19.9 Side slope. The truck, with gross load, shall be driven in both directions on a 20-percent side slope, and the ability to start and idle the truck engine while the truck is parked in both directions on the side slope shall be determined. Engine oil and fuel pressures shall be noted. There shall be no evidence of loss of fuel, lubricants, or fire fighting agents.

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4.6.19.10 Endurance. The truck shall be operated over the following roads under conditions and for distances specified:

a. Paved roads at an average speed of 45 mph - 600 miles fully loaded, plus 600 miles with agent tanks filled to one-half capacity, plus 600 miles unloaded except for driver, fuels, and lubricants

b. Graded gravel roads at an average speed of 25 mph - 100 miles fully loaded, plus 100 miles with agent tanks filled to one-half capacity, plus 100 miles unloaded except for driver, fuels, and lubricants

c. Belgian block roads at an average speed of 20 mph - 100 miles fully loaded, plus 100 miles with agent tanks filled to one-half capacity, plus 100 miles unloaded except for driver, fuels, and lubricants

During the 1,800 miles of paved road operation, the truck shall be accelerated from a standstill to its maximum rated speed after each 100 miles. At the completion of the 1,800-mile paved road operation, the engine shall be operated at idle for 4 hours. The truck shall be checked for faulty operation, misalignment, and damage throughout all the tests. The distance traveled in driving to the Aberdeen Proving Ground may be used to satisfy part of the paved road phase, as applicable, providing complete records are kept during the trip.

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be in accordance with the instructions of the procuring activity.

6. NOTES

6.1 Intended use. The A/S32P-12 truck is intended to be used as an air transportable structural fire fighting vehicle and for use at military installations world wide for protection of real material and property.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification

b. Whether a group 1 (gasoline) or group 2 (diesel) chassis is to be provided (see 3.7)

c. When communication equipment is to be provided (see 3.7.2.4)

d. When nondirectional cross-country (NDCC) tires are to be provided (see 3.7.6)

e. When winterization for -40°F operation is to be provided (see 3.13.2)

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f. When the truck is to be undercoated (see 3.17.1)

g. Lettering and numbering required on the truck (see 3.18.1)

h. Location and conditions for preproduction testing (see 4.4).

6.3 Definitions

6.3.1 Automatic transmission. Automatic transmission is one in which all combinations of variable torque ratio changes can be accomplished by the operator with one setting of the shift control and without more than an instantaneous interruption of the power flow.

6.3.2 Semiautomatic, continuous drive, automatically stepped. Semiautomatic drive, automatically stepped transmission is one in which a portion of the total torque conversion ratios is continuously variable, but in which the remaining ratio conversions are accomplished through range control by the use of automatically stepped gear ratios which do not require more than an instantaneous power flow interruption with each selected range.

6.4 Additional aircraft information. Additional information on the Type C-130 aircraft and the A/A32H-4 dual rail cargo handling system is contained in the following USAF cargo aircraft technical orders:

a. T. O. 1C-130A-9 Cargo Loading
C-130A, B, D, E, RC-130A, HC-130B

b. T. O. 1C-130A-9D Supplement to the above

c. T. O. 1C-130(H) H-9 Loading, Delivery, and Recovery Instructions HC-130H

(Copies of the above technical orders should be requested from the contracting officer.)

Custodian:
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